

ConstructionSceneReady: Evidence-grounded compilation and task-scene validation for BIM-driven robotic assembly

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Abstract

Robot-ready construction simulation requires more than converting building geometry into a simulation format. We ask which building information actually supports a robotic construction task in simulation, and how its absence, error, or imprecision changes task outcomes. ConstructionSceneReady compiles IFC source into task-activated multi-scale OpenUSD scenes and validates them against an explicit task-scene contract with evidence-bounded repair. On 105 fault-injection cases the contract recovers all 141 findings without false positives and repairs 97.1% to pristine equivalence; an agent constrained to the same whitelist preserves it, while an unconstrained agent produces incorrect-but-passing scenes in 22.7%. Paired humanoid episodes show that *wrong* symbolic and referential values are discretely fatal while a *missing* semantic label still degrades outcomes. The 12 mm reward criterion also masks anchor-relative interface errors: geometrically valid insertion collapses to 2.0% at a 10 mm hole-coordinate error while reward success stays $\sim 80\%$, a blind spot the contract closes with two independently verified rules.

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Declaration of competing interest

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